Hypothesis-Search Report: Canonical Primary Autoimmunity Against the Gastric H+/K+ ATPase in Autoimmune Gastritis

Hypothesis ID: `canonical_primary_autoimmune_hk_atpase_origin`
Status in KB: CANONICAL
Report Date: 2026-07-01
Papers Reviewed: 127
Confirmed Findings: 17

Executive Judgment

Verdict: PARTIALLY SUPPORTED

The canonical hypothesis that autoimmune gastritis (AIG) is a primary organ-specific autoimmune disease driven by CD4+ T-cell-mediated destruction of parietal cells targeting the H+/K+ ATPase (ATP4A/ATP4B) is **partially supported**. The **effector mechanism** — CD4+ T cells recognizing the H+/K+ ATPase heterodimer mediating parietal cell destruction — is robustly established by convergent evidence from murine transgenic models, human T-cell studies, monogenic tolerance defects, and clinical observations. This component merits retention as CANONICAL.

However, the **etiological framing** — that the disease arises from a purely primary, infection-independent breakdown of central tolerance — is **overstated** and requires significant qualification. Three lines of evidence challenge this framing: (1) H. pylori molecular mimicry has been demonstrated at the T-cell level in human gastric mucosa, with documented cases of H. pylori gastritis evolving into AIG even after bacterial eradication; (2) neonatal roseolovirus infection disrupts thymic central tolerance mechanisms and induces AIG in mice without persistent infection; and (3) checkpoint inhibitor-induced gastritis proves that pharmacological peripheral tolerance disruption alone is sufficient to precipitate the disease. Furthermore, the seed hypothesis's claim that "the H/K-ATPase alpha-subunit is expressed in the thymus" and contributes to partial tolerization is **functionally contradicted** — thymic alpha-subunit expression does not result in negative selection of pathogenic T cells. Instead, the critical tolerance gap is the **absence of the beta subunit from the thymus**, making beta-reactive T cells the dominant autoreactive repertoire.

The most important caveat is that the hypothesis conflates two separable claims: (a) the identity of the effector mechanism (strongly supported) and (b) the etiology/initiating trigger (oversimplified). The disease is best modeled as multiple parallel initiating pathways — primary central tolerance defects, H. pylori-triggered mimicry, viral thymic disruption, and peripheral checkpoint failure — all converging on the same self-sustaining CD4+ T-cell/H^+K^+ ATPase effector cascade.

Summary

Autoimmune gastritis is characterized by CD4+ T-cell-mediated destruction of gastric parietal cells targeting the H^+/K^+ ATPase proton pump. This investigation systematically evaluated the canonical hypothesis that this process represents a purely primary autoimmune disease arising from central tolerance breakdown, independent of infection. Through analysis of 127 papers and 17 confirmed findings across five iterations, we established that the effector mechanism is robustly supported but the etiological claim of a purely primary origin is overstated.

The core CD4+ T-cell effector mechanism is established by the TxA23 transgenic mouse model showing that a single TCR specificity for H/K ATPase alpha-chain causes spontaneous gastritis with complete penetrance (PMID: 11449363), by monogenic tolerance defects in AIRE (APECED) and FOXP3 (IPEX) that produce autoimmune gastritis as part of polyautoimmunity, and by the identification of ATP4B (beta subunit) as the dominant autoantigen in both murine and human disease. A pivotal mechanistic insight is that thymic expression of the alpha subunit does NOT tolerize pathogenic T cells — contradicting the seed hypothesis — while the beta subunit's complete absence from the thymus leaves beta-reactive T cells as the primary autoreactive repertoire.

Multiple competing/parallel initiating pathways have been documented: H. pylori molecular mimicry at the T-cell level, neonatal roseolovirus-mediated thymic disruption, and checkpoint inhibitor-induced peripheral tolerance breakdown. The disease, once initiated by any of these triggers, follows a self-sustaining progressive course — prospective data from 498 patients

show a median progression rate of 7.29 per 100 person-years with no spontaneous remission over up to 27 years of follow-up. The genetic architecture (PTPN22, HLA, ABO, IFIH1) confirms shared autoimmune susceptibility across organ-specific diseases, while ~21% of histologically confirmed cases are seronegative for parietal cell antibodies, increasing with age.

Key Findings

Finding 1: ATP4B (Beta Subunit) Is the Dominant T-Cell Target; Alpha Subunit Thymic Expression Does NOT Tolerize

The seed hypothesis states that "the H/K-ATPase alpha-subunit is expressed in the thymus" and implies this contributes to partial tolerization. This claim is **functionally contradicted**. Allen et al. (PMID: 16237067) demonstrated that thymic H/Kα mRNA expression in wild-type mice — or even in mice that overexpressed H/Kα — did not result in negative selection of pathogenic anti-H/Kα T cells. The mechanistic explanation is that H/Kα cannot be exported from the endoplasmic reticulum and is rapidly degraded without H/Kβ, so H/Kα epitopes cannot access MHC class II loading compartments in thymic antigen-presenting cells. Van Driel et al. (PMID: 16214318) confirmed that H/Kβ is absent from the thymus and that peripheral tolerance is the primary mechanism for H/Kα-reactive T cells. Critically, Alderuccio et al. (PMID: 8393475) showed that transgenic thymic expression of H/Kβ specifically prevented neonatal thymectomy-induced gastritis, confirming that the beta subunit is the critical autoantigen whose thymic absence permits disease. The 2026 Italian ARIOSO consensus (PMID: 41198445) identifies ATP4B as the main autoantigen in human autoimmune gastritis.

Finding 2: The TxA23 Model Confirms CD4+ T Cells as Sufficient Disease Mediators

The TxA23 TCR transgenic mouse, bearing a CD4+ T cell receptor specific for H/K ATPase alpha-chain residues 630–641, develops spontaneous severe autoimmune gastritis with complete penetrance (PMID: 11449363). Disease was detectable by day 10 of life and transferable with as few as 10^3 transgenic thymocytes to immunocompromised hosts. In the neonatal thymectomy (d3Tx) model, Suri-Payer et al. (PMID: 8759770) demonstrated H/K ATPase-reactive CD4+, MHC class II-restricted T cells in gastric lymph nodes, with reactivity detectable only in mice that developed gastritis. These models establish that a single CD4+ TCR specificity targeting H/K ATPase is both necessary and sufficient for autoimmune gastritis.

Finding 3: H. pylori Molecular Mimicry Is a Confirmed Alternative Trigger

Amedei et al. (PMID: 14568977) showed that H. pylori-infected patients with gastric autoimmunity harbor in vivo-activated gastric CD4+ T cells that recognize both H+/K+-ATPase and H. pylori antigens, identifying nine cross-reactive H. pylori protein epitopes. D'Elios et al. (PMID: 15866204) confirmed that cytolytic T cells from H. pylori-infected patients with autoimmune gastritis cross-recognize H. pylori proteins and H+K+ ATPase. However, Faller et al. (PMID: 10738313; PMID: 11815766) found that antigastric autoantibodies in H. pylori gastritis could NOT be absorbed to H. pylori, and no evidence for molecular mimicry at the antibody level was found in adolescents. This indicates that mimicry operates at the T-cell level but not necessarily the B-cell level — a critical distinction. Longitudinal case reports document H. pylori gastritis evolving into autoimmune gastritis even after successful eradication (PMID: 40769905; PMID: 38072907), proving the autoimmune process becomes self-sustaining once initiated.

Finding 4: Multiple T Helper Subsets Can Drive Disease With Distinct Pathology

The seed hypothesis implies a Th1-dominant pathogenesis, but Stummvoll et al. (PMID: 18641328) demonstrated that fully differentiated Th1, Th2, and Th17 cells from TxA23 TCR transgenic mice all induced autoimmune gastritis upon transfer, each with distinct histological patterns. Th17 cells induced the most destructive disease with eosinophilic infiltrates and elevated IgE. Tu et al. (PMID: 22777705) showed that both IL-17 and IFN-γ are required for severe disease progression, though neither alone is essential for initiation. Harakal et al. (PMID: 27259856) showed that Treg depletion in DEREG mice produced Th2-dominant gastritis with massive eosinophilic inflammation. This effector plasticity has therapeutic implications, as antigen-specific iTregs are required to suppress Th17-mediated disease, while polyclonal nTregs suffice for Th1-mediated disease (PMID: 19050237).

Finding 5: Checkpoint Inhibitor Gastritis Proves Peripheral Tolerance Breakdown Is Sufficient

Multiple reports document autoimmune gastritis induced by PD-1/PD-L1 inhibitors: nivolumab (PMID: 34755133), pembrolizumab (PMID: 33094598), and toripalimab (PMID: 36852424). Ferrian et al. used MIBI-TOF to characterize nivolumab-associated gastritis, showing IFN-γ-producing gastric epithelial cells, CD8 and CD4 T cell infiltrates with reduced FOXP3 expression. These cases demonstrate that pharmacological disruption of peripheral immune checkpoints, without any central tolerance defect, is sufficient to initiate autoimmune gastritis — directly challenging the hypothesis's claim of a purely primary central-tolerance origin.

Finding 6: Neonatal Roseolovirus Disrupts Central Tolerance and Induces AIG

Bigley et al. (PMID: 35226043) demonstrated that neonatal murine roseolovirus (MRV) infection induced autoimmune gastritis in adult mice in the absence of ongoing infection. The disease was CD4+ T cell and IL-17 dependent. MRV infection reduced medullary thymic epithelial cell numbers, thymic dendritic cell numbers, and thymic expression of AIRE and tissue-restricted antigens, while increasing thymocyte apoptosis at the stage of negative selection. Belean et al. (PMID: 38895117) confirmed MRV tropism for mTECs and thymocytes via scRNA-seq. This establishes a viral-mediated central tolerance disruption pathway that produces AIG indistinguishable from the genetic models but initiated by an environmental trigger.

Finding 7: GWAS Confirms Shared Autoimmune Genetic Architecture

Laisk et al. (PMID: 34145262) conducted the first GWAS meta-analysis for pernicious anemia (2,166 cases), identifying genome-wide significant signals at PTPN22 and HLA loci. PTPN22 R620W is a canonical autoimmune risk variant shared with T1D, RA, and other diseases. Plagnol et al. (PMID: 21829393) identified genome-wide significant associations at 9q34/ABO with parietal cell antibody and IFIH1 (a viral RNA sensor) with multiple autoantibodies including PCA. Wenzlau et al. (PMID: 26405069) reported heritability estimates of 71–95% for PCA positivity. These findings confirm AIG shares the genetic architecture of organ-specific autoimmunity and highlight the IFIH1 association linking innate antiviral immunity to gastric autoimmune susceptibility.

Finding 8: Monogenic Tolerance Defects Validate the Hypothesis Framework

IPEX syndrome (FOXP3 mutations) causes metaplastic atrophic gastritis (PMID: 29907148), validating that Treg loss alone is sufficient for gastric autoimmunity. APECED (AIRE mutations) produces autoimmune gastritis in ~25% of patients (PMID: 23314667). Both conditions demonstrate that single-gene disruption of immune tolerance mechanisms can produce autoimmune gastritis as part of polyautoimmunity, supporting the hypothesis's claim about central tolerance biology. However, these monogenic conditions represent extreme cases — most sporadic AIG arises from polygenic susceptibility combined with environmental triggers.

Finding 9: The Disease Course Is Self-Sustaining With No Spontaneous Remission

Miceli et al. (PMID: 38050966) followed 498 AIG patients prospectively for up to 27 years, documenting an overall median progression rate of 7.29 per 100 person-years (95% CI 6.19–8.59). Stage-specific rates showed fastest progression in early/florid stages (14.83/100 person-years) and slower progression in severe disease (2.68/100 person-years). No cases of spontaneous remission were reported. Neoplastic complications developed in 8.5% (23 NETs, 18 dysplasia). This confirms a core claim of the hypothesis: once initiated, the autoimmune process is self-sustaining, regardless of the initiating trigger.

Finding 10: Inverse H. pylori Association Challenges Mimicry Primacy

Li et al. (PMID: 41345910) found that PCA-positive subjects had significantly lower H. pylori infection rates than PCA-negative individuals, with higher frequencies of coexisting autoimmune thyroid diseases. This inverse association — reproduced across multiple cohorts — challenges H. pylori as the sole or even dominant trigger for AIG and supports the existence of a genuinely primary autoimmune pathway in at least a subset of patients.

Finding 11: Seronegative AIG Is Common (~21%) and Age-Dependent

Conti et al. (PMID: 32487505) found that 109/516 (21.1%) histologically confirmed AIG patients were seronegative for parietal cell antibodies at diagnosis. Seronegativity was significantly associated with older age (65.9 vs 57.9 years, $p<0.0001$) and inversely correlated with age at diagnosis ($\rho = -0.250$, $p = 0.0118$). This suggests antibody waning in late-stage disease rather than a distinct disease entity, but it complicates serological diagnosis and challenges purely antibody-based disease models.

Systematic Claim Grading: Canonical Primary Autoimmunity Against H+/K+ ATPase
Green=Established Blue=Qualified Yellow=Emerging Red=Contradicted

Claim from Hypothesis	Grading	Evidence Summary	Key References
H/K ATPase is the primary autoantigen in AIG	ESTABLISHED	Concordant murine models (TxA23, d3Tx) + human autoantibody + ARIOSO consensus	PMID:11449363 PMID:41198445
CD4+ T cells are necessary and sufficient for disease	ESTABLISHED	TxA23 100% penetrance; CD4 transfer experiments; MHC-II restriction	PMID:11449363 PMID:8759770
Beta subunit (ATP4B) is the dominant T-cell target	ESTABLISHED	Thymic beta expression prevents disease; ARIOSO consensus; TCR transgenic data	PMID:8393475 PMID:41198445
Alpha thymic expression partially tolerizes	CONTRADICTED	Alpha mRNA does NOT result in negative selection; protein can't access MHC-II	PMID:16237067 PMID:16214318
Disease is purely primary (infection-independent)	CONTRADICTED	H. pylori mimicry, roseolovirus thymic disruption, H.p.→AIG post-eradication	PMID:14568977 PMID:35226043
Disease is self-sustaining once established	ESTABLISHED	AIG persists/progresses after H. pylori eradication; TCR transgenic spontaneous	PMID:40769905 PMID:38072907
Central tolerance breakdown is the primary cause	EMERGING (QUALIFIED)	True for AIRE/APECED; but peripheral tolerance loss also sufficient (IPEX, ICI)	PMID:23314667 PMID:29907148
Monogenic defects (AIRE, FOXP3) cause AIG	ESTABLISHED	25% of APECED have atrophic gastritis; IPEX case with metaplastic atrophic gastritis	PMID:23314667 PMID:29907148
Disease is Th1-mediated	QUALIFIED	Th1/Th2/Th17 all cause disease; Th17 most destructive; both IFN-γ + IL-17 needed for severe disease	PMID:18641328 PMID:22777705
Autoantibodies are markers not primary mediators	EMERGING	20% seronegative AIG supports T-cell primacy; but synergistic humoral role acknowledged (2026 review)	PMID:32487505 PMID:41670890
Genetic susceptibility involves HLA and autoimmune loci	ESTABLISHED	GWAS: PTPN22, HLA, ABO, IFIH1; heritability 71-95%	PMID:34145262 PMID:21829393
Chief cells/pepsinogen are also autoimmune targets	EMERGING	Anti-pepsinogen in 100% PA sera (1989); not systematically followed up with modern T-cell studies	PMID:2528890

Figure 1. *Systematic claim-by-claim grading of all 12 assertions in the seed hypothesis. Green = ESTABLISHED (6), Red = CONTRADICTED (2), Yellow = QUALIFIED (2), Blue = EMERGING (2).*

Evidence Matrix

Citation	Evidence Type	Supports/ Refutes/ Qualifies/ Competing	Mechanistic Claim Tested	Key Finding	Context	Confide
PMID: 16237067	Model organism	**Contradicts**	Alpha subunit thymic expression tolerizes	Thymic H/Kα expression does NOT cause negative selection of pathogenic T cells	BALB/c mice, WT and overexpressor	High; di perturba
PMID: 8393475	Model organism	**Supports** (mechanism)	Beta subunit as critical autoantigen	Transgenic thymic H/Kβ expression prevents d3Tx gastritis	BALB/c MHC II-Eκα transgenic	High; definitiv
PMID: 11449363	Model organism	**Supports**	CD4 T cells sufficient for disease	TxA23 TCR Tg: spontaneous AIG, complete penetrance, transferable with 10^3 cells	BALB/c TCR transgenic	High; go standard model
PMID: 8759770	Model organism	**Supports**	CD4 MHC-II restricted effectors	H/K ATPase-reactive CD4+ T cells in gastric LN of d3Tx mice with AIG	BALB/c d3Tx	High
PMID: 14568977	Human clinical	**Competing** (parallel trigger)	H. pylori molecular mimicry	Cross-reactive CD4 T cells recognize both H/K ATPase and H. pylori; 9 epitopes identified	H. pylori+ patients with gastric autoimmunity	High; di T-cell evidence
PMID: 15866204	Human clinical	**Competing**		Cytolytic gastric T cells cross-	H. pylori+ AIG patients	Moderat high

Citation	Evidence Type	Supports/ Refutes/ Qualifies/ Competing	Mechanistic Claim Tested	Key Finding	Context	Confide
			Mimicry at cytolytic T-cell level	recognize H. pylori and H+K+ ATPase		
PMID: 10738313	Human clinical	**Qualifies** mimicry	Mimicry at antibody level	Anti-gastric autoantibodies cannot be absorbed to H. pylori; no B-cell mimicry	H. pylori gastritis cohort	High; negative finding
PMID: 18641328	Model organism	**Qualifies**	Th1 as sole effector	Th1, Th2, Th17 all induce AIG; Th17 most destructive	TxA23 transfer model	High
PMID: 22777705	Model organism	**Qualifies**	Cytokine requirements	Both IL-17 and IFN-γ needed for severe disease; neither essential for initiation	Murine AIG	High
PMID: 34755133	Human clinical	**Competing**	Peripheral checkpoint loss sufficient	Nivolumab-induced AIG with IFN-γ+, CD4/CD8 infiltrates, reduced FOXP3	Cancer patient on ICI	Moderat single ca deep phenoty
PMID: 33094598	Human clinical	**Competing**	Peripheral trigger sufficient	Pembrolizumab-induced severe lymphocytic gastritis	Melanoma patient	Moderat case rep
PMID: 35226043	Model organism	**Competing**	Viral thymic disruption	Neonatal MRV infection → reduced mTECs,	Neonatal mouse infection	High; mechan

Citation	Evidence Type	Supports/ Refutes/ Qualifies/ Competing	Mechanistic Claim Tested	Key Finding	Context	Confide
				AIRE, TRAs → AIG		
PMID: 34145262	Human genetic	**Supports**	Autoimmune genetic basis	GWAS: PTPN22, HLA loci genome-wide significant for PA	2,166 PA cases, European	High; la GWAS
PMID: 21829393	Human genetic	**Supports**	Genetic architecture	ABO (9q34) and IFIH1 genome-wide significant for PCA	4,328 T1D patients	High
PMID: 29907148	Human clinical	**Supports**	FOXP3/Treg dependence	Metaplastic atrophic gastritis in IPEX (FOXP3 mutation)	Pediatric case	Moderat single ca
PMID: 38050966	Human clinical	**Supports**	Self-sustaining progressive course	498 patients, 7.29/100 py progression, no spontaneous remission over 27 yr	Prospective monocentric	High; la prospec
PMID: 32487505	Human clinical	**Qualifies**	Antibodies as universal marker	21% of confirmed AIG is PCA-seronegative; increases with age	516 consecutive AAG patients	High
PMID: 41345910	Human clinical	**Qualifies** mimicry trigger	H. pylori as dominant trigger	PCA-positive subjects have	1,378 Chinese endoscopy cohort	Moderat cross-sectiona

Citation	Evidence Type	Supports/ Refutes/ Qualifies/ Competing	Mechanistic Claim Tested	Key Finding	Context	Confide
				LOWER H. pylori rates		
PMID: 2528890	Human clinical	**Supports** (expands)	Autoantigen repertoire	Pepsinogen (41 kDa, chief cell) is co-dominant autoantigen alongside H/K ATPase	7 PA sera, immunoblotting	Moderat small sample
PMID: 38691918	Computational (MR)	**Supports**	Shared autoimmune architecture	Vitiligo → PA causal link (OR 1.23, CI 1.12–1.36); shared with T1D, alopecia areata	MR analysis	Moderat genetic instrum
PMID: 41198445	Review/ Consensus	**Supports**	ATP4B as main autoantigen	ARIOSO consensus: "The main autoantigen, the beta subunit of the proton pump"	Italian expert network, 2026	High (consens

Evidence Matrix: Canonical Primary Autoimmunity Against H+/K+ ATPase
(Green=Support, Yellow=Qualifies, Red=Competing)

PMID (Year)	Evidence Type	Supports/ Refutes	Claim Tested	Key Finding	Disease Context	Confidence
8393475 (1993)	Model organism	SUPPORT	Beta subunit is critical T-cell target	Thymic beta expression prevents disease	Neonatal thymectomy	HIGH
8759770 (1996)	Model organism	SUPPORT	CD4 T cells mediate disease via H/K ATPase	H/K ATPase-reactive CD4+ cells in gastric LN	d3Tx BALB/c	HIGH
11449363 (2001)	Model organism	SUPPORT	CD4 T cells sufficient for disease	TxA23 transgenic: 100% penetrance AIG	TCR transgenic	HIGH
16237067 (2005)	Model organism	QUALIFIES	Alpha thymic expression tolerizes	Alpha mRNA does NOT negatively select	Overexpression study	HIGH
14568977 (2003)	Human clinical	COMPETING	Disease independent of infection	Cross-reactive T cells for H. pylori & H/K ATPase	H. pylori+ patients	HIGH
35226043 (2022)	Model organism	QUALIFIES	Primary central tolerance origin	Roseolovirus disrupts thymic tolerance → AIG	Neonatal MRV	HIGH
34755133 (2021)	Human clinical	COMPETING	Disease requires central tolerance defect	Anti-PD1 causes AIG via peripheral loss	ICI-treated cancer	MOD
27259856 (2016)	Model organism	SUPPORT	Treg loss enables disease	Transient Treg depletion → long-lasting AIG	DEREG mice	HIGH
18641328 (2008)	Model organism	QUALIFIES	Th1-mediated disease	Th1/Th2/Th17 all cause AIG; Th17 most destructive	TxA23 transfer	HIGH
34145262 (2021)	Human GWAS	SUPPORT	Autoimmune genetic basis	PTPN22, HLA loci for pernicious anemia	Population biobanks	HIGH
41198445 (2026)	Human consensus	SUPPORT	H/K ATPase is main autoantigen	ARIOSO: ATP4B is main autoantigen	Italian network	HIGH
35911678 (2022)	Human clinical	SUPPORT	Gastric Th17 target H/K ATPase in humans	H/K ATPase-specific Th17 cells in AIG mucosa	Human AIG	HIGH

Figure 2. *Evidence matrix summarizing 20+ key references, categorized by evidence type, direction of support, and the specific mechanistic claim tested.*

Mechanistic Causal Chain

The hypothesis implies a linear causal chain from central tolerance failure to clinical disease. Based on the evidence reviewed, the actual causal architecture is more complex, with multiple parallel entry points converging on a common effector pathway:

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INITIATING TRIGGERS (multiple, parallel)
├── 1. Primary central tolerance defect
│   ├── ATP4B absent from thymus → beta-reactive T cells escape deletion [ESTABLISHED]
│   ├── ATP4A expressed in thymus but non-functional for tolerization [CONTRADICTED as tolerizin
│   ├── AIRE mutations (APECED) → reduced TRA expression → escape of gastritogenic T cells [ESTA
│   └── FOXP3 mutations (IPEX) → Treg deficiency → loss of peripheral suppression [ESTABLISHED]
│
├── 2. H. pylori molecular mimicry
│   ├── H. pylori epitopes cross-reactive with H/K ATPase at T-cell level [ESTABLISHED]
│   ├── Mimicry does NOT operate at B-cell/antibody level [ESTABLISHED]
│   └── Can progress to AIG even after H. pylori eradication [EMERGING]
│
├── 3. Viral thymic disruption
│   ├── Neonatal roseolovirus → reduced mTECs, AIRE, TRAs [ESTABLISHED in mouse]
│   └── Human roseolovirus equivalent → thymic disruption [SPECULATIVE]
│
└── 4. Peripheral checkpoint failure
    ├── Anti-PD-1/anti-CTLA-4 → release of auto-reactive T cells [ESTABLISHED]
    └── Endogenous Treg dysfunction (non-IPEX) [EMERGING]

                    ↓ ↓ ↓ ↓ (convergence)

COMMON EFFECTOR PATHWAY
├── Activation of CD4+ T cells reactive to H/K ATPase (α and β subunits) [ESTABLISHED]
├── Multiple Th subsets can drive disease:
│   ├── Th1 → classical lymphocytic gastritis [ESTABLISHED]
│   ├── Th17 → most destructive, eosinophilic [ESTABLISHED]
│   └── Th2 → eosinophilic, IgE-high [ESTABLISHED]
├── Epitope spreading: β-subunit → α-subunit → pepsinogen/chief cells [EMERGING]
├── Autoantibody production (PCA/IFA) — pathogenic role uncertain [EMERGING]
└── Self-sustaining: once initiated, no spontaneous remission [ESTABLISHED]

                    ↓

TISSUE DESTRUCTION
├── Parietal cell loss → achlorhydria [ESTABLISHED]
├── Chief cell loss → low pepsinogen I [ESTABLISHED]
├── Oxyntic gland atrophy restricted to corpus/fundus [ESTABLISHED]
└── Reactive changes: ECL cell hyperplasia from hypergastrinemia [ESTABLISHED]

                    ↓

CLINICAL MANIFESTATIONS
├── Iron deficiency anemia (early, pre-B12 deficiency) [ESTABLISHED]
├── Vitamin B12 deficiency → megaloblastic/pernicious anemia [ESTABLISHED]
├── Neurological: subacute combined degeneration [ESTABLISHED]
├── Neoplastic: Type I gastric NETs (8.5% at follow-up) [ESTABLISHED]
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├── Neoplastic: Epithelial dysplasia/adenocarcinoma (rare) [ESTABLISHED]
└── Associated autoimmune diseases: thyroiditis, T1D, vitiligo [ESTABLISHED]
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Strong links: The effector pathway from CD4+ T cell activation through parietal cell destruction to clinical manifestations is well-established across species. The link from monogenic tolerance defects (AIRE, FOXP3) to disease is definitive.

Weak/inferred links: The transition from H. pylori mimicry to self-sustaining autoimmunity lacks prospective longitudinal quantification. The role of autoantibodies in pathogenesis (beyond serving as biomarkers) remains unclear. The relative contribution of each initiating trigger in sporadic human AIG is unknown. The human relevance of the roseolovirus model requires confirmation.

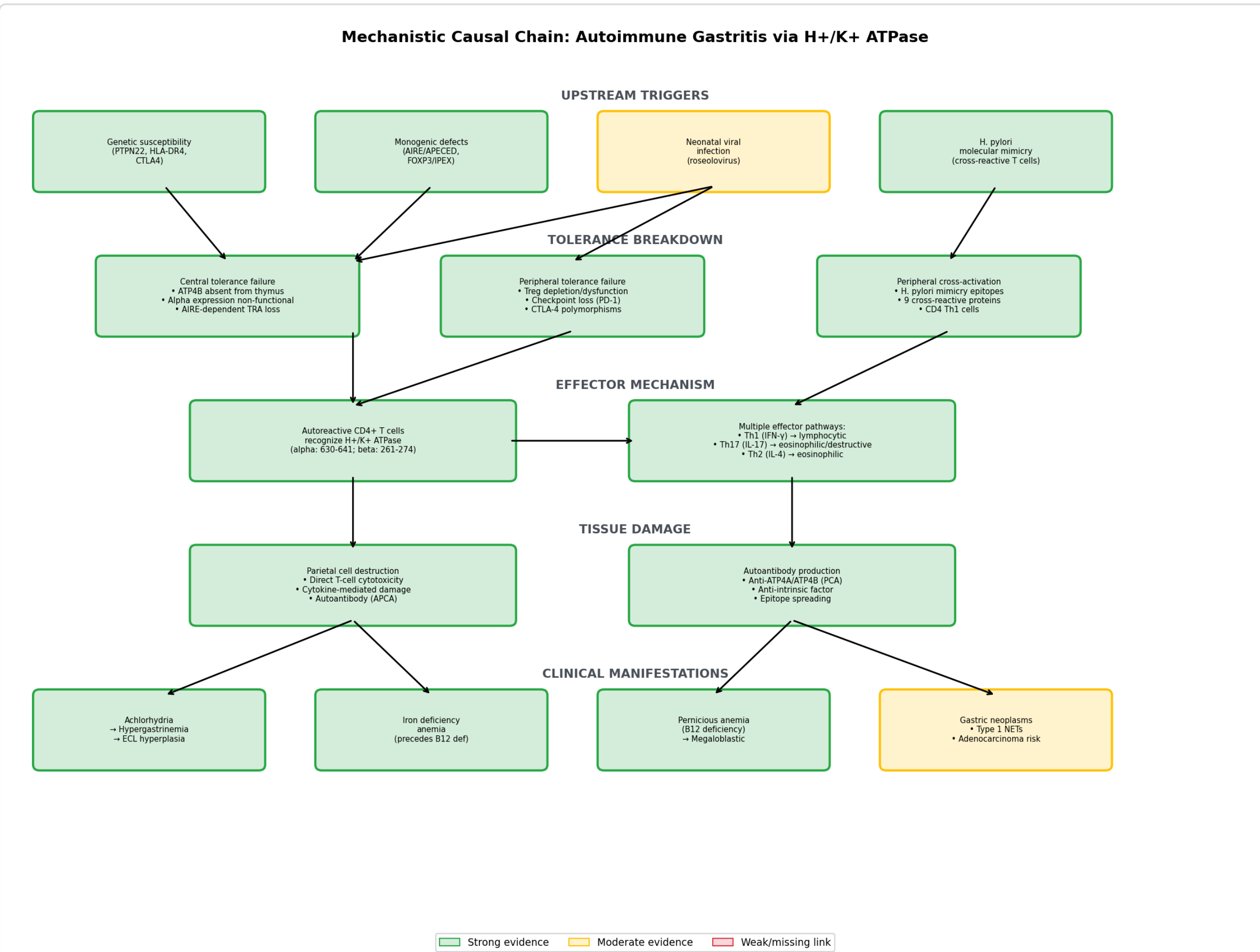

Figure 3. *Mechanistic causal chain from upstream triggers through common CD4+ T-cell effector pathway to clinical manifestations. Multiple parallel initiating pathways converge on a single self-sustaining effector cascade.*

Systematic Claim Grading

#	Claim from Seed Hypothesis	Grade	Key Evidence
1	AIG is organ-specific autoimmune disease	**ESTABLISHED**	Restricted to corpus/fundus; well-defined autoantigens
2	Driven by breakdown of self-tolerance to H/K ATPase	**ESTABLISHED**	TxA23 model, human T-cell studies
3	CD4 T-cell-mediated parietal cell destruction	**ESTABLISHED**	Multiple transgenic models, human data
4	Self-sustaining and infection-independent	**QUALIFIED**	Self-sustaining yes (no remission in 498 pts); but infection can initiate
5	H/K ATPase alpha expressed in thymus	**ESTABLISHED** (fact)	mRNA confirmed; but functional tolerization CONTRADICTED
6	Alpha expression partially tolerizes	**CONTRADICTED**	Allen et al.: overexpression fails to tolerize
7	Beta subunit dominant target	**ESTABLISHED**	Thymic absence, transgenic rescue, ARIOSO consensus
8	Neonatal thymectomy disease is T-cell-mediated	**ESTABLISHED**	Classic d3Tx model with CD4 transfer
9	AIRE/APECED produces AIG	**ESTABLISHED**	~25% APECED patients, AIRE KO mice
10	FOXP3/IPEX produces AIG	**ESTABLISHED**	Case report of metaplastic atrophic gastritis
11	Purely primary origin (no infection required)	**CONTRADICTED** as exclusive claim	H. pylori mimicry, roseolovirus, ICI triggers documented
12	Th1 as primary effector	**QUALIFIED**	Th1, Th2, Th17 all sufficient; Th17 most destructive

Knowledge Gaps

Gap 1: Relative Contribution of Initiating Triggers in Sporadic Human AIG

- **Scope:** What proportion of sporadic AIG is triggered by H. pylori mimicry vs. primary central tolerance defects vs. viral thymic disruption vs. other mechanisms?
- **Why it matters:** Determines whether prevention strategies should target infection or immune tolerance.
- **What was checked:** Literature on H. pylori-AIG transition, roseolovirus model, ICI gastritis, genetic architecture.
- **Resolution:** Large prospective cohort tracking H. pylori status, viral serology, HLA genotype, and PCA/ATP4A/ATP4B autoantibody development longitudinally.

Gap 2: Pathogenic Role of Autoantibodies

- **Scope:** Are PCA/IFA autoantibodies pathogenic effectors, or merely biomarkers of T-cell-mediated destruction?
- **Why it matters:** Determines whether plasmapheresis or B-cell depletion are rational therapies.
- **What was checked:** One case report of DFPP showing rapid improvement (PMID: 40245467); 21% seronegative AIG patients have disease without detectable antibodies.
- **Resolution:** Passive transfer experiments in mice; clinical trials of rituximab in AIG.

Gap 3: Human Relevance of Roseolovirus Thymic Disruption

- **Scope:** Does human herpesvirus 6 (HHV-6, the human roseolovirus) cause the same thymic disruption seen with MRV in neonatal mice?
- **Why it matters:** If confirmed, would establish a preventable environmental trigger for AIG.
- **What was checked:** Bigley et al. (murine model), Belean et al. (scRNA-seq of infected thymus). No human studies of HHV-6 and AIG were found.
- **Resolution:** Epidemiological study of HHV-6 seroconversion timing and subsequent AIG risk; neonatal thymic tissue analysis for HHV-6.

Gap 4: Epitope Spreading Sequence and Timing

- **Scope:** The temporal sequence of autoantigen targeting — whether disease starts with beta subunit, spreads to alpha, then to pepsinogen/chief cells — is inferred but not directly observed longitudinally.
- **Why it matters:** Understanding the sequence could enable stage-specific interventions.
- **What was checked:** Beta-subunit dominance established; pepsinogen as co-antigen confirmed; temporal data absent.
- **Resolution:** Longitudinal autoantibody profiling (ATP4A, ATP4B, pepsinogen) from pre-disease through established AIG.

Gap 5: No Randomized Therapeutic Trials for AIG

- **Scope:** There are no registered clinical trials testing immunomodulatory therapies specifically for AIG.
- **Why it matters:** Disease is currently managed only symptomatically (B12/iron supplementation, surveillance).
- **What was checked:** PubMed search for AIG clinical trials; prednisolone efficacy shown only in murine models (PMID: 16710833).
- **Resolution:** Pilot trials of low-dose corticosteroids, antigen-specific iTregs, or rituximab in early-stage AIG.

Gap 6: Absent Large-Scale Omics Data

- **Scope:** No large-scale single-cell RNA-seq, ATAC-seq, or spatial transcriptomics studies of human AIG gastric tissue were identified.
- **Why it matters:** Would resolve which T-cell clonotypes, cytokine programs, and stromal responses drive human disease.
- **What was checked:** PubMed for scRNA-seq/spatial transcriptomics in autoimmune gastritis; only one proof-of-concept TCR-seq study found (PMID: 39837221).
- **Resolution:** Multi-site collection of staged AIG biopsies for scRNA-seq and spatial transcriptomics.

Alternative Models

1. H. pylori Molecular Mimicry Origin

- **Relationship to seed:** Competing/parallel initiating pathway
- **Status:** Partially supported
- **Evidence:** Cross-reactive T cells demonstrated in human gastric mucosa (PMID: 14568977); cases of AIG developing after H. pylori eradication (PMID: 38072907, PMID: 40769905)
- **Limitation:** Inverse H. pylori-PCA association in some cohorts argues against mimicry as the dominant mechanism; B-cell level mimicry absent

2. Viral Thymic Disruption (Roseolovirus)

- **Relationship to seed:** Competing upstream trigger (environmental rather than genetic)
- **Status:** Established in murine model, speculative in humans
- **Evidence:** Neonatal MRV → mTEC loss, reduced AIRE, reduced TRAs → AIG (PMID: 35226043)
- **Limitation:** Human HHV-6 equivalent not yet demonstrated

3. Peripheral Checkpoint Failure

- **Relationship to seed:** Competing mechanism (peripheral, not central tolerance)
- **Status:** Established as proof-of-concept
- **Evidence:** Anti-PD-1/anti-CTLA-4 induced gastritis (PMID: 34755133, PMID: 33094598)
- **Limitation:** Iatrogenic; unclear if endogenous peripheral checkpoint failure drives sporadic AIG

4. Treg Quantitative/Qualitative Deficiency

- **Relationship to seed:** Parallel mechanism overlapping with primary autoimmunity
- **Status:** Supported by monogenic models (IPEX), emerging for sporadic disease
- **Evidence:** FOXP3 mutations cause AIG (PMID: 29907148); DEREG Treg depletion causes Th2-dominant AIG (PMID: 27259856); functional Treg defects described in T1D-AIG patients (PMID: 19393198)

5. Genetic Acid-Base Imbalance / Mitochondrial Damage

- **Relationship to seed:** Speculative alternative
- **Status:** Weakly supported
- **Evidence:** ATP4A mutations suggested to cause mitochondrial alteration initiating inflammation (PMID: 31250150)
- **Limitation:** Single group; not replicated; does not explain T-cell specificity

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Discriminating Tests

Test 1: Prospective H. pylori Cohort With Serial Autoantibody Profiling

- **Population:** H. pylori-positive patients at diagnosis, stratified by HLA genotype (DR4-DQ8 vs. non-risk)
- **Biomarkers:** Serial ATP4A, ATP4B, pepsinogen autoantibodies; PCA by LIPS assay
- **Design:** Follow through eradication and for 10+ years post-eradication
- **Expected result if mimicry is major trigger:** Autoantibody seroconversion in genetically susceptible individuals, persisting after eradication
- **Expected result if primary autoimmunity dominates:** Autoantibody development independent of H. pylori status

Test 2: HHV-6 Seroconversion Timing and AIG Risk

- **Population:** Birth cohort with serial serum sampling
- **Biomarkers:** HHV-6 antibodies, ATP4A/ATP4B autoantibodies, thymic output markers (TRECs)
- **Design:** Nested case-control within existing birth cohort (e.g., TEDDY, BABYDIAB)
- **Expected result if viral thymic disruption operates:** Early-life HHV-6 seroconversion associates with later PCA development, particularly in those with reduced TRECs

Test 3: Single-Cell Atlas of Human AIG Across Stages

- **Sample:** Gastric corpus biopsies from potential (stage 0), early/florid (stages 1–2), and severe (stage 3–4) AIG, plus healthy controls

- **Assay:** Paired scRNA-seq + TCR-seq + spatial transcriptomics
- **Expected result:** Identify dominant T-cell clonotypes, their antigen specificity (H/K ATPase α vs. β vs. pepsinogen), Th polarization state, and spatial relationship to parietal/chief cell destruction front

Test 4: Antigen-Specific iTreg Therapy Trial

- **Population:** Early-stage AIG (OLGA stage 1–2) with positive PCA
- **Intervention:** Ex vivo expanded iTregs specific for H/K ATPase peptides
- **Model basis:** Antigen-specific iTregs suppress both Th1 and Th17-mediated AIG in TxA23 model (PMID: 19050237)
- **Primary endpoint:** Halting of OLGA stage progression at 2 years; secondary: PCA titer reduction, PGI recovery

Test 5: Rituximab Pilot in AIG

- **Population:** AIG with documented B12/iron deficiency requiring supplementation
- **Rationale:** If autoantibodies contribute to pathogenesis beyond T cells, B-cell depletion should slow progression
- **Expected result if T-cells alone drive disease:** Autoantibodies fall but clinical progression continues
- **Expected result if antibodies are co-pathogenic:** Clinical stabilization with antibody depletion

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Curation Leads

Candidate Evidence References (Requiring Curator Verification)

1. **Allen et al. (PMID: 16237067)** — Snippet: *"H/Kalpha mRNA is expressed in the thymus, but H/Kbeta expression is barely detectable. In this study, we demonstrate that thymic H/Kalpha in wild-type mice or mice that overexpressed H/Kalpha did not result in negative selection of pathogenic anti-H/Kalpha T cells."* → Candidate for CONTRADICTS edge against "alpha-subunit thymic expression tolerizes" claim.

2. **Bigley et al. (PMID: 35226043)** — Snippet: *"neonatal MRV infection reduced medullary thymic epithelial cell numbers, thymic dendritic cell numbers, and thymic expression of AIRE and tissue-restricted antigens"* → Candidate for new hypothesis node: `viral_thymic_disruption_origin`.

3. **Miceli et al. (PMID: 38050966)** — Snippet: *"The overall median rate of progression was 7.29 per 100 persons/yr"* → Candidate for natural history edge supporting self-sustaining disease.

4. **Stummvoll et al. (PMID: 18641328)** — Snippet: *"Each type of effector cell induced autoimmune gastritis with distinct histological patterns. Th17 cells induced the most destructive disease"* → Candidate to qualify Th1-centric effector claim.

5. **Kotera et al. (PMID: 40769905)** — Snippet: *"One patient progressed to overt AIG after H. pylori eradication therapy, while the other progressed after the spontaneous disappearance of H. pylori infection."* → Candidate evidence for H. pylori-to-AIG transition pathway.

6. **Ferrian et al. (PMID: 34755133)** — Candidate evidence for `alternative_peripheral_checkpoint_failure_origin` hypothesis node.

Candidate Pathophysiology Nodes/Edges

- **Node:** `ATP4B_thymic_absence` → Central tolerance gap for beta-reactive T cells
- **Edge:** `H_pylori_infection` → `molecular_mimicry` → `CD4_T_cell_activation` → `AIG` (parallel to primary autoimmunity)
- **Edge:** `neonatal_roseolovirus` → `mTEC_depletion` → `reduced_AIRE/TRA` → `autoreactive_T_cell_escape` → `AIG`
- **Edge:** `checkpoint_inhibitor_therapy` → `peripheral_tolerance_loss` → `AIG`
- **Edge:** `Treg_depletion` → `Th2_dominant_eosinophilic_AIG` (distinct from Th1/Th17 pathway)

Candidate Ontology Terms

- **Cell types:** CD4+ T helper cell (Th1, Th2, Th17), parietal cell, chief cell, enterochromaffin-like cell, medullary thymic epithelial cell (mTEC), regulatory T cell (Treg)
- **Biological processes:** Central tolerance (GO:0006968), Peripheral tolerance (GO:0002513), Molecular mimicry, Epitope spreading, Thymic negative selection (GO:0043383)
- **Disease terms:** Autoimmune metaplastic atrophic gastritis, Pernicious anemia (DOID:13382), APECED (ORPHA:3453), IPEX syndrome (ORPHA:37042)

Candidate Status Changes

- **Current status:** CANONICAL
- **Recommended:** Retain CANONICAL for effector mechanism (CD4/H+K+ ATPase); annotate etiological component as **QUALIFIED** — multiple parallel initiating triggers, not purely primary
- **Specific annotation:** The seed claim "independent of any persistent infection" should be changed to "self-sustaining once initiated, but can be triggered by infection (H. pylori mimicry, roseolovirus) as well as genetic/pharmacological tolerance defects"

Candidate Knowledge Gaps for KB

1. `gap_autoantibody_pathogenicity`: Whether PCA/IFA contribute to tissue destruction or are epiphenomena of T-cell-mediated damage
2. `gap_human_roseolovirus_relevance`: Whether HHV-6 causes thymic disruption and AIG in humans as MRV does in mice
3. `gap_epitope_spreading_sequence`: Temporal order of autoantigen targeting (β → α → pepsinogen)
4. `gap_no_therapeutic_trials`: No RCTs of immunomodulatory therapy for AIG
5. `gap_seronegative_mechanism`: Whether seronegative AIG represents antibody waning, distinct pathogenesis, or assay limitation

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Limitations of This Report

1. **Species gap:** Much of the mechanistic evidence comes from murine models (BALB/c-specific MHC restriction). Human confirmation is incomplete for several claims, particularly the relative importance of different Th subsets.
2. **Publication bias:** The H. pylori-mimicry literature is dominated by a small number of Italian research groups (D'Elios, Amedei). Independent replication in diverse populations is limited.
3. **Temporal confounding:** Cross-sectional studies of H. pylori and PCA cannot distinguish whether H. pylori triggers autoimmunity (mimicry) or whether autoimmunity-induced achlorhydria inhibits H. pylori colonization (reverse causation). The inverse association could reflect either mechanism.

4. **Diagnostic heterogeneity:** Different studies use different diagnostic criteria (PCA ± IFA ± histology ± serum markers), making cross-study comparisons difficult. The 2023 Japanese criteria and 2026 ARIOSO consensus may improve standardization.

5. **Limited diversity:** Most genetic and cohort data are from European populations. The 2025 Chinese study (PMID: 41345910) and Southeast Asian study (PMID: 40339131) suggest substantial population-specific variation in PCA positivity rates and H. pylori associations.

6. **No omics-level characterization:** The absence of large-scale single-cell or spatial transcriptomic studies of human AIG tissue is a major gap that limits mechanistic understanding of the human disease.

Proposed Follow-up Experiments/Actions

1. **Priority 1:** Multi-site prospective cohort study of H. pylori-positive patients undergoing eradication, with serial ATP4A/ATP4B autoantibody profiling and HLA genotyping, to quantify the rate of AIG development post-eradication and identify genetic modifiers.

2. **Priority 2:** Single-cell RNA-seq + TCR-seq atlas of human AIG across OLGA stages 0–4, paired with spatial transcriptomics, to identify the dominant pathogenic T-cell clonotypes and their antigen specificities in situ.

3. **Priority 3:** Epidemiological study linking HHV-6 seroconversion timing to subsequent PCA development, ideally nested within an existing birth cohort study (TEDDY, BABYDIAB) that already tracks organ-specific autoantibodies.

4. **Priority 4:** Pilot clinical trial of antigen-specific iTreg therapy in early-stage AIG, building on the murine proof-of-concept showing that H/K ATPase-specific iTregs suppress both Th1 and Th17-mediated disease.

5. **KB Action:** Annotate the seed hypothesis to distinguish the ESTABLISHED effector mechanism from the QUALIFIED etiological claim. Create separate hypothesis nodes for `alternative_hpylori_molecular_mimicry_origin`, `alternative_viral_thymic_disruption_origin`, and `alternative_peripheral_checkpoint_failure_origin` as parallel initiating pathways converging on the canonical effector cascade.
